

DRAVYANTRA: ENTERPRISE TELEMETRICS & FLEET INTELLIGENCE PLATFORM

Comprehensive Technical Architecture, Mathematical Computation Engine, and Operations Manual

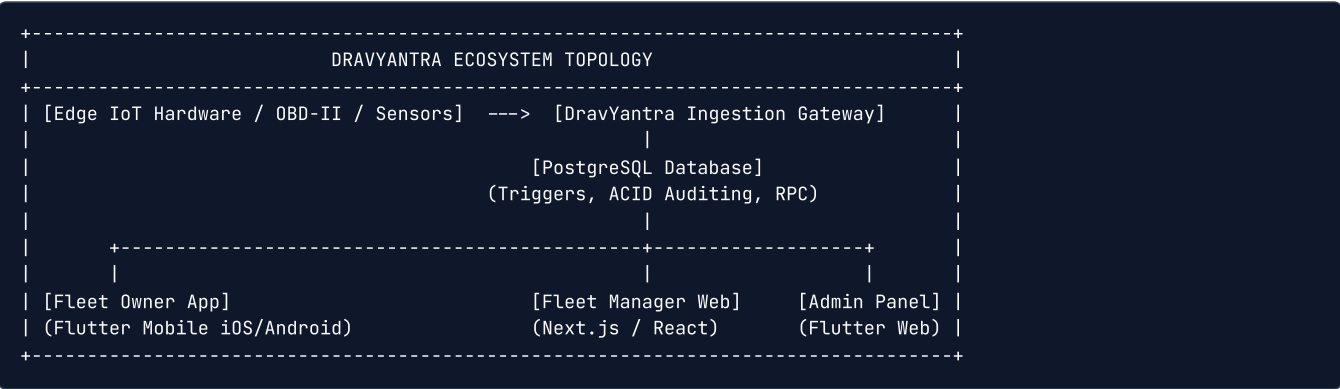
TABLE OF CONTENTS

1. [Executive Platform Overview](#)
2. [Technology Stack & Architectural Rationale](#)
 - 2.1 [DravYantra Fleet Owner Mobile App \(Frontend\)](#)
 - 2.2 [DravYantra Fleet Manager Web Dashboard](#)
 - 2.3 [DravYantra Admin & Super-Admin Dashboard](#)
 - 2.4 [DravYantra Core Backend & Telemetry Ingestion Pipeline](#)
 - 2.5 [Database Architecture & Data Persistence Layer](#)
3. [DravYantra Backend: Calculation Engines, Algorithms & Mathematical Formulas](#)
 - 3.1 [7-Layer Telemetry Ingestion Pipeline Architecture](#)
 - 3.2 [GPS Distance Calculation & Anti-Jitter Filtering \(Haversine Formula\)](#)
 - 3.3 [Dynamic Speed-to-Mileage Degradation Engine](#)
 - 3.4 [Fuel Consumption & Real-Time Fuel Used Calculation](#)
 - 3.5 [Fuel Saved & Verified Money Saved Engine](#)
 - 3.6 [Excess Speeding Fuel Wasted & Financial Loss Calculation](#)
 - 3.7 [Idling Fuel Consumption, Idling Cost & Alert Thresholds](#)
 - 3.8 [Real-Time Fuel Theft Detection Engine](#)
 - 3.9 [Automated Fuel Refill Detection Engine](#)
 - 3.10 [ESG Carbon \(CO2\) Emissions, CO2 Avoided & Carbon \(C\) Reduction Engine](#)
 - 3.11 [Live Trip Safety Score Calculation](#)
 - 3.12 [Exposure-Normalized Driver Scorecard Engine](#)
 - 3.13 [Normalized Vehicle Health Score Engine](#)
 - 3.14 [Audited Savings Wallet & 7-Day Baseline Learning Engine](#)
 - 3.15 [Vehicle Lifetime Statistics Aggregation Engine](#)
4. [DravYantra Fleet Manager Dashboard: Exhaustive Feature Breakdown](#)
 - 4.1 [Executive Overview & Fleet KPI Dashboard](#)
 - 4.2 [Live Fleet GIS Tracking & Interactive Map](#)
 - 4.3 [Trips Management, Dispatch & Route Analysis](#)
 - 4.4 [Fuel Intelligence, Theft Audits & Refills](#)
 - 4.5 [Audited Savings Wallet & ROI Analytics](#)
 - 4.6 [Carbon ESG & Sustainability Reporting](#)
 - 4.7 [Driver Scorecards, Safety Leaderboard & Behavior](#)
 - 4.8 [Vehicle Health, Diagnostics & Maintenance Schedules](#)
 - 4.9 [Alerts, Tampering & Real-Time Incident Watchtower](#)
 - 4.10 [Automated Reports Generation & Data Export Hub](#)
 - 4.11 [Fleet Organization Settings & Threshold Calibration](#)
 - 4.12 [Support & In-App Helpdesk](#)
5. [DravYantra Super-Admin Dashboard: Platform Operations & Administration](#)
 - 5.1 [Super-Admin Role & Global Responsibilities](#)
 - 5.2 [Platform Operations Center & System Health Metrics](#)
 - 5.3 [Fleet Organizations & Onboarding Approval Lifecycle](#)

- 5.4 [IoT Hardware & Telemetry Device Inventory Management](#)
 - 5.5 [Global Vehicle & Asset Registry](#)
 - 5.6 [Global Trip Surveillance & Incident Investigation](#)
 - 5.7 [Subscription Plans, Billing & Commercial Engine](#)
 - 5.8 [Global User Management & Role-Based Access Control \(RBAC\)](#)
 - 5.9 [Platform-Wide Security Alerts & Tampering Watchtower](#)
 - 5.10 [Audit Trails, Security Logging & Compliance Tracker](#)
 - 5.11 [Global Support Helpdesk & SLA Management](#)
 - 5.12 [Recycle Bin & 30-Day Soft Deletion Recovery Engine](#)
6. [Summary Reference Table of Key Backend Constants & Formulas](#)

1. EXECUTIVE PLATFORM OVERVIEW

DravYantra is an enterprise-grade IoT Telematics, Fleet Intelligence, and Audited Fuel Accounting Platform. The system bridges edge hardware (GPS trackers, fuel level sensors, vibration sensors, CAN-bus / OBD-II readers) with cloud computing to give commercial fleet operators 100% visibility over fuel economics, driver behavior, carbon footprint (ESG), and vehicle mechanical integrity.



2. TECHNOLOGY STACK & ARCHITECTURAL RATIONALE

System Component	Technology Stack Chosen	Key Architectural Rationale
Fleet Owner Mobile App	Flutter (Dart), Firebase Auth, Google Maps SDK, Provider/Riverpod	Native 60 FPS mobile performance, single iOS/Android codebase, offline caching
Fleet Manager Web Dashboard	Next.js 14 (App Router), React 18, TypeScript, Tailwind CSS, Recharts	Server-side rendering (SSR), high-speed real-time GIS mapping, modular analytics
Admin / Super-Admin Panel	Flutter Web & Desktop, Material 3, Riverpod, REST Client, Charts	Uniform desktop/web multi-pane layout, robust enterprise RBAC, audit logging
Backend & Ingestion API	Node.js, Express.js (ES6+), PG Connection Pool, Nodemailer	High-throughput event-loop I/O for IoT, atomic transactions, sub-second telemetry
Database & Persistence	PostgreSQL 15+, PL/pgSQL Triggers, JSONB, Connection Pooling	ACID financial-grade audit trails, strict relational schemas, DB-level triggers

2.1 DravYantra Fleet Owner Mobile App (Frontend)

- **Technology:** Flutter Framework (Dart language), Material Design 3, Google Maps SDK for Flutter, Firebase Authentication & Cloud Messaging (FCM).
- **Why this stack:**

1. **Single Codebase for iOS & Android:** Fleets employ drivers and managers using diverse mobile devices. Flutter provides true native rendering across both ecosystems with zero UI deviation.
2. **High-Performance Canvas Rendering:** Telemetry updates (speed, fuel gauge, live map marker transitions) require steady 60–120 FPS rendering without lag.
3. **Instant Push Notifications:** Firebase Cloud Messaging enables immediate background notifications for fuel theft, speed violations, and unauthorized vehicle movement.

2.2 DravYantra Fleet Manager Web Dashboard

- **Technology:** Next.js 14+ (App Router), React 18, TypeScript, Tailwind CSS, Lucide Icons, Recharts, OpenStreetMap / Leaflet / Mapbox.
- **Why this stack:**
 1. **Speed & Hybrid Server-Side Rendering (SSR):** Provides instant initial page load speeds and optimizations for enterprise portals.
 2. **Strict Type Safety with TypeScript:** Large data structures (telemetry packets, multi-variable trip aggregates, audit reports) require compile-time type verification to prevent runtime UI bugs.
 3. **Modular Component Architecture:** Allows rapid rendering of live charts, telemetry cards, and multi-vehicle map layers.

2.3 DravYantra Admin & Super-Admin Dashboard

- **Technology:** Flutter Web & Desktop, Material Design 3, Riverpod State Management, REST API integration.
- **Why this stack:**
 1. **Desktop & Web Multi-Pane Layout:** The super-admin console requires expansive data tables, real-time platform monitoring cards, and hardware provisioning consoles.
 2. **Consistent State Architecture:** Riverpod guarantees predictable state updates across subscription management, device inventory, and organization onboarding.

2.4 DravYantra Core Backend & Telemetry Ingestion Pipeline

- **Technology:** Node.js, Express.js REST Framework, PostgreSQL Client (`pg` pool), Firebase Admin SDK.
- **Why this stack:**
 1. **Non-Blocking Asynchronous I/O:** IoT devices send high-frequency telemetry every 5–10 seconds. Node.js processes thousands of concurrent HTTP POST / Webhook telemetry packets per second without thread exhaustion.
 2. **Modular Service Layer:** The engine is split into specialized decoupled micro-services (`gpsEngine` , `fuelEngine` , `idleEngine` , `safetyEngine` , `tripCalculationEngine` , `savingsWalletEngine`).

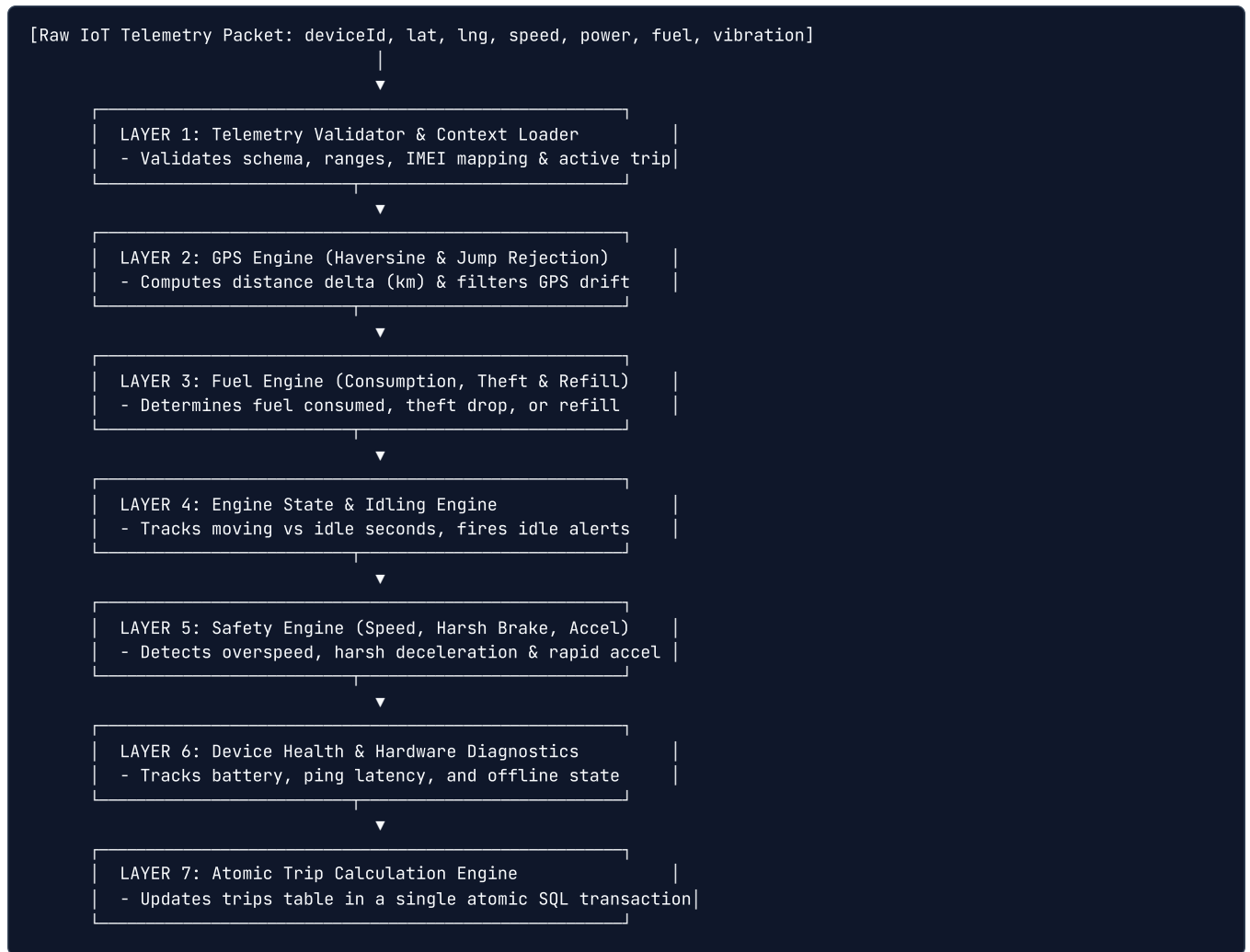
2.5 Database Architecture & Data Persistence Layer

- **Technology:** PostgreSQL 15+ with Relational Tables, JSONB Documents, Foreign Key Cascades, and PL/pgSQL Triggers (`compute_idle_money_wasted`).
- **Why this stack:**
 1. **ACID Financial Integrity:** Fuel accounting and savings calculations require strict ACID transactions so that money saved and losses are never miscalculated during concurrent writes.
 2. **Dual-Layer Fallback via Database Triggers:** In case an API update bypasses the service layer, PostgreSQL database triggers automatically re-calculate fuel consumption and money wasted on the raw row level.

3. DRAVYANTRA BACKEND: CALCULATION ENGINES, ALGORITHMS & MATHEMATICAL FORMULAS

The DravYantra backend features an end-to-end mathematical engine that converts raw telemetry pings into financial, operational, and ESG insights.

3.17-Layer Telemetry Ingestion Pipeline Architecture



3.2 GPS Distance Calculation & Anti-Jitter Filtering (Haversine Formula)

A. Mathematical Formula

To compute the exact spherical distance between two sequential GPS coordinates $(\text{lat}_1, \text{lng}_1)$ and $(\text{lat}_2, \text{lng}_2)$ on Earth:

$$\Delta\sigma = 2 \cdot \arcsin \left(\sqrt{\sin^2 \left(\frac{\Delta\phi}{2} \right) + \cos(\phi_1) \cos(\phi_2) \sin^2 \left(\frac{\Delta\lambda}{2} \right)} \right)$$
$$\text{Distance Delta } (\Delta D) = R \cdot \Delta\sigma$$

Where:

- ϕ_1, ϕ_2 = Latitude of previous point and current point converted to Radians: $\phi = \text{lat} \cdot \frac{\pi}{180}$
- $\Delta\phi = (\phi_2 - \phi_1)$ = Difference in Latitude in Radians
- $\Delta\lambda = (\lambda_2 - \lambda_1)$ = Difference in Longitude in Radians
- R = Earth's mean radius = **6371.0 km**

B. Anti-Drift & Anti-Jump Filter Rules

```
IF power == false OR speed == 0 THEN
    distanceDelta = 0.0 km
ELSE IF rawDistance < gpsDriftThresholdKm (0.010 km = 10 meters) THEN
    distanceDelta = 0.0 km -- Filter out GPS satellite jitter
ELSE IF rawDistance > gpsMaxJumpKm (5.0 km per 5s packet) THEN
    distanceDelta = 0.0 km -- Reject teleportation / faulty GPS jump
ELSE
    distanceDelta = rawDistance
END IF
```

C. Numerical Example

- Point 1: \$(19.0760, 72.8777)\$ (Mumbai)
- Point 2: \$(19.0800, 72.8800)\$
- Raw calculated Haversine distance = \$0.505\text{ km}\$ (\$505\text{ meters}\$)
- Condition: Speed = \$60\text{ km/h}\$, Engine = ON \$\rightarrow\$ Valid movement recorded: \$+0.505\text{ km}\$ added to trip distance.

3.3 Dynamic Speed-to-Mileage Degradation Engine

Heavy commercial vehicles experience exponential aerodynamic drag and engine friction at higher speeds. DravYantra calculates dynamic mileage from real-time speed.

A. Speed-to-Mileage Degradation Table

Speed Range	Mileage (km/L)	Efficiency Gain / Loss	Engine Operational Status
0 - 39 km/h	4.00 km/L	0.0% (Baseline)	Urban / Low gear operation
40 - 59 km/h	5.00 km/L	+25.0% (Best Economy)	Optimal Eco-Cruising Zone
60 - 69 km/h	4.00 km/L	0.0% (Baseline)	Standard Highway Baseline
70 - 79 km/h	3.60 km/L	-10.0% Degradation	Mild Speed Penalty
80 - 89 km/h	3.40 km/L	-15.0% Degradation	Moderate Aerodynamic Loss
90 - 99 km/h	3.20 km/L	-20.0% Degradation	Severe Fuel Waste Zone
100 - 109 km/h	3.00 km/L	-25.0% Degradation	High Speed Critical Penalty
110 - 119 km/h	2.80 km/L	-30.0% Degradation	Excessive Commercial Stress
\$\ge\$ 120 km/h	2.60 km/L	-35.0% Degradation	Worst Efficiency Danger Zone

B. Mathematical Formula

Current Mileage (M_{current}) = {5.00, if $40 \leq v \leq 59$ 4.00, if $v < 40$ or $(60 \leq v < 70)$ $4.00 \cdot (1 - \text{Degradation Percentage})$, if $v \geq 70$

Where:

- v = Current vehicle telemetry speed in km/h .
- M_{baseline} = Standard vehicle baseline efficiency (4.00 km/L default, or customized per organization).

3.4 Fuel Consumption & Real-Time Fuel Used Calculation

A. Mathematical Formula

Fuel Consumed (F_{used}) = $\frac{\text{Distance Traveled (D)}}{M_{\text{current}}}$

B. Variable Definitions:

- F_{used} : Cumulative fuel burned over distance D , measured in **Liters (L)**.
- D : Cumulative distance traveled, measured in **Kilometers (km)**.

- M_{current} : Dynamic speed-adjusted mileage, measured in km/L .

C. Step-by-Step Numerical Example

- A truck covers a trip segment of $D = 400 \text{ km}$ at a steady speed of $v = 80 \text{ km/h}$.
- From the speed degradation table, at 80 km/h , $M_{\text{current}} = 3.40 \text{ km/L}$.
- Calculation:

$$F_{\text{used}} = \frac{400}{3.40} = 117.65 \text{ Liters}$$

3.5 Fuel Saved & Verified Money Saved Engine

When a driver operates within the eco-speed window ($40 - 59 \text{ km/h}$), the vehicle achieves 5.0 km/L instead of baseline 4.0 km/L , saving fuel.

A. Mathematical Formula

$$\text{Expected Fuel at Baseline } (F_{\text{expected}}) = \frac{D}{M_{\text{baseline}}}$$

$$\text{Fuel Saved } (F_{\text{saved}}) = \max(0, F_{\text{expected}} - F_{\text{used}})$$

$$\text{Money Saved (INR ₹)} = F_{\text{saved}} \times P_{\text{fuel}}$$

B. Variable Definitions:

- F_{expected} : Fuel that would have been consumed at baseline efficiency (4.0 km/L).
- F_{used} : Actual fuel consumed at current dynamic efficiency (5.0 km/L).
- F_{saved} : Net fuel saved in **Liters**.
- P_{fuel} : Fuel price per liter (e.g., $\text{₹}95.00/\text{L}$ or configured fleet price).
- Money Saved : Direct financial profit generated by fuel-efficient driving.

C. Step-by-Step Numerical Example

- A truck drives $D = 400 \text{ km}$ at 50 km/h ($M_{\text{current}} = 5.0 \text{ km/L}$, $M_{\text{baseline}} = 4.0 \text{ km/L}$, $P_{\text{fuel}} = \text{₹}95/\text{L}$).
- $F_{\text{expected}} = \frac{400}{4.0} = 100.00 \text{ Liters}$
- $F_{\text{used}} = \frac{400}{5.0} = 80.00 \text{ Liters}$
- $F_{\text{saved}} = 100.00 - 80.00 = \mathbf{20.00 \text{ Liters}}$
- $\text{Money Saved} = 20.00 \text{ L} \times \text{₹}95 = \mathbf{\text{₹}1,900.00 \text{ Saved}}$

3.6 Excess Speeding Fuel Wasted & Financial Loss Calculation

Driving above baseline highway speeds burns excess diesel due to aerodynamic resistance.

A. Mathematical Formula

$$\text{Speeding Fuel Wasted } (F_{\text{wasted_speed}}) = \max(0, F_{\text{used}} - F_{\text{expected}})$$

$$\text{Speeding Financial Loss (INR ₹)} = F_{\text{wasted_speed}} \times P_{\text{fuel}}$$

B. Step-by-Step Numerical Example

- A truck drives $D = 400 \text{ km}$ recklessly at $v = 120 \text{ km/h}$ ($M_{\text{current}} = 2.60 \text{ km/L}$, $M_{\text{baseline}} = 4.0 \text{ km/L}$, $P_{\text{fuel}} = \text{₹}95/\text{L}$).
- $F_{\text{expected}} = \frac{400}{4.0} = 100.00 \text{ Liters}$
- $F_{\text{used}} = \frac{400}{2.60} = 153.85 \text{ Liters}$
- $F_{\text{wasted_speed}} = 153.85 - 100.00 = \mathbf{53.85 \text{ Liters wasted}}$
- $\text{Speeding Financial Loss} = 53.85 \text{ L} \times \text{₹}95 = \mathbf{\text{₹}5,115.75 \text{ Lost}}$

3.7 Idling Fuel Consumption, Idling Cost & Alert Thresholds

An engine left running while stationary consumes fuel without generating mileage, causing wear and carbon emissions.

A. Mathematical Formula

$$\text{Idle Money Wasted (INR ₹)} = T_{\text{idle_seconds}} \times R_{\text{idle_rate}}$$

$$\text{Idle Fuel Wasted (} F_{\text{idle_liters}} \text{)} = \frac{\text{Idle Money Wasted}}{P_{\text{fuel}}}$$

B. Variable Definitions:

- $T_{\text{idle_seconds}}$: Cumulative stationary seconds where Engine Power is **ON** and Speed is **\$0\text{ km/h}** \$.
- $R_{\text{idle_rate}}$: Baseline financial burn rate = **\$₹1.70\text{ per second}** \$ (standard heavy diesel idle rate embedded in database triggers and analytics).
- P_{fuel} : Fuel price per liter (e.g., $\text{\$₹95.00/L}$).

C. Step-by-Step Numerical Example

- A vehicle idles at a loading dock for $T_{\text{idle}} = 20\text{ minutes} = 1,200\text{ seconds}$ ($P_{\text{fuel}} = \text{\$₹95/L}$).
- $\text{Idle Money Wasted} = 1,200\text{ sec} \times \text{\$₹1.70/sec} = \text{\$₹2,040.00}$
- $\text{Idle Fuel Wasted} = \frac{\text{\$₹2,040}}{\text{\$₹95}} = \text{\$₹21.47 Liters}$

D. Automated Multi-Tier Idling Alert Triggers

- Idle Warning Alert:** Triggered when continuous idling reaches 300 seconds (5 minutes).
- Idle Critical Alert:** Triggered when continuous idling reaches 900 seconds (15 minutes).

3.8 Real-Time Fuel Theft Detection Engine

Fuel siphoning or theft occurs when fuel levels drop sharply while the engine is turned off or while the vehicle is stationary.

A. Detection Conditions Logic

```
A Fuel Theft Event is triggered IF AND ONLY IF:  
1. Engine Power State = OFF (power == false)  
2. Vehicle Speed = 0 km/h (isStationary == true)  
3. Time Delta between packets ≤ 5 seconds (Window)  
4. (Previous_Fuel - Current_Fuel) ≥ Fuel_Theft_Threshold (Default: 0.70 Liters)
```

B. Mathematical Formula

$$\Delta F = F_{\text{previous}} - F_{\text{current}}$$

$$\text{Theft Loss (} F_{\text{theft}} \text{)} = \{\Delta F, \text{ if Engine is OFF, Speed} = 0, \text{ and } \Delta F \geq 0.70 \text{ L}, \text{ otherwise}$$

$$\text{Theft Financial Loss (INR ₹)} = F_{\text{theft}} \times P_{\text{fuel}}$$

C. Step-by-Step Numerical Example

- A parked vehicle has $F_{\text{previous}} = 180.0\text{ L}$.
- In the next 5-second packet with engine OFF, $F_{\text{current}} = 145.0\text{ L}$.
- $\Delta F = 180.0 - 145.0 = 35.0\text{ L} \geq 0.70\text{ L}$ threshold.
- Result: **Instant Critical Theft Alert** dispatched with GPS coordinates.
- $\text{Theft Financial Loss} = 35.0\text{ L} \times \text{\$₹95/L} = \text{\$₹3,325.00}$.

3.9 Automated Fuel Refill Detection Engine

When fuel levels increase significantly, DravYantra logs an audited refill event to verify fuel bills and eliminate phantom billing.

A. Detection Logic & Formula

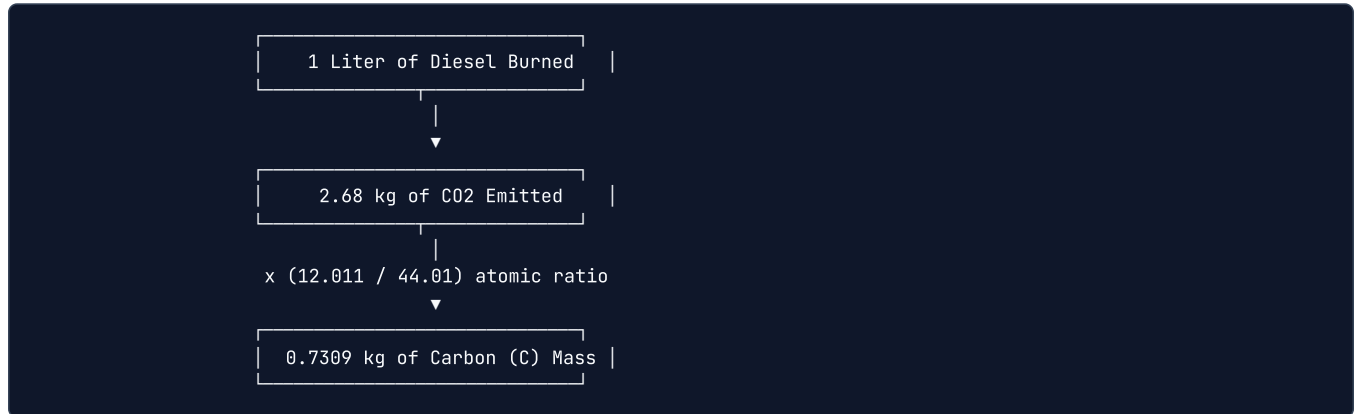
$$\Delta F_{\text{increase}} = F_{\text{current}} - F_{\text{previous}}$$

$$\text{Refill Event Triggered} \iff \Delta F_{\text{increase}} \geq \text{Refill Threshold (5.00 Liters)}$$

- Stored attributes: `vehicle_plate`, `trip_id`, `fuel_before`, `fuel_after`, `amount_filled`, `lat`, `lng`, `timestamp`.

3.10 ESG Carbon (CO₂) Emissions, CO₂ Avoided & Carbon (C) Reduction Engine

DravYantra provides audited ESG metrics adhering to international GHG (Greenhouse Gas) Protocol standards.



A. Mathematical Formulas

1. Gross Carbon Dioxide (\$CO₂\$) Emitted:

$$\text{Gross } CO_2 \text{ Emitted (kg } CO_2) = F_{\text{used}} \times 2.68 \text{ kg } CO_2/\text{Liter}$$

2. Carbon Dioxide (\$CO₂\$) Avoided / Saved:

$$\text{Net } CO_2 \text{ Avoided (kg } CO_2) = F_{\text{saved}} \times 2.68 \text{ kg } CO_2/\text{Liter}$$

3. Pure Elemental Carbon (\$C\$) Reduced:

$$\text{Elemental Carbon (C) Mass Reduced (kg C)} = \text{Net } CO_2 \text{ Avoided} \times \left(\frac{12.011}{44.01} \right) = \text{Net } CO_2 \text{ Avoided} \times 0.2727$$

Where:

- \$2.68\$: Official combustion emission factor for commercial diesel (in $\text{kg } CO_2/\text{per Liter}$).
- \$12.011\$: Atomic mass of pure Carbon (\$C\$).
- \$44.01\$: Molecular mass of Carbon Dioxide (\$CO_2\$).

B. Step-by-Step Numerical Example

- A fleet optimizes driver speeds over a month, achieving $F_{\text{saved}} = 1,500.00 \text{ Liters}$ of diesel saved.
- \$CO₂\$ Avoided:**

$$\text{Net } CO_2 \text{ Avoided} = 1,500 \times 2.68 = \mathbf{4,020.00 \text{ kg } CO_2 (4.02 \text{ Metric Tonnes)}}$$

- Carbon (\$C\$) Mass Reduced:**

$$\text{Carbon Reduced} = 4,020 \times \left(\frac{12}{44} \right) = \mathbf{1,096.36 \text{ kg of Pure Carbon Reduced}}$$

3.11 Live Trip Safety Score Calculation

Computed in real-time on every incoming telemetry packet for active trips.

A. Mathematical Formula

$$\text{Trip Score} = \max \left(0, ; \min \left(100, ; 100 - (N_{os} \times 2) - (N_{hb} \times 3) - (N_{ra} \times 2) - P_{alert} \right) \right)$$

Where:

- N_{os} = Total count of overspeeding events ($v > 80 \text{ km/h}$). Each subtracts **\$2\text{ points}**.
- N_{hb} = Total count of harsh braking events ($\Delta v \geq 40 \text{ km/h drop in 5s}$). Each subtracts **\$3\text{ points}**.
- N_{ra} = Total count of rapid acceleration events ($\Delta v \geq 25 \text{ km/h gain in 5s}$). Each subtracts **\$2\text{ points}**.
- P_{alert} = Excessive alert penalty: **\$5\text{ points}** if total alerts > 5 , else \$0.

3.12 Exposure-Normalized Driver Scorecard Engine

Driver safety evaluation must be fair: a driver driving $2,000 \text{ km}$ naturally encounters more situations than a driver driving 10 km . DravYantra normalizes all penalties per **\$100\text{ km}** of exposure.

DRIVER SCORECARD COMPUTATION MATRIX			
Safety Dimension	Rate Metric	Multiplier	Max Penalty Cap
Overspeeding	Events / 100 km	x 5.0	-25 Points Max
Excess Idling	Idle Hours / 100 km	x 4.0	-25 Points Max
Fuel Inefficiency	(1 - Eff_Ratio)	x 40.0	-25 Points Max
Rash Driving	Harsh Evts / 100 km	x 5.0	-25 Points Max
FINAL DRIVER SCORE	100 - (P _{os} + P _{idle} + P _{eff} + P _{rash})		[0 - 100 Points]

A. Mathematical Formulas

1. Exposure Distance:

$$D_{\text{exp}} = \max(1.0, ; \text{Total Distance in km})$$

2. Overspeeding Penalty (P_{os}):

$$\text{Overspeed Rate } (R_{\text{os}}) = \left(\frac{N_{\text{speed_events}}}{D_{\text{exp}}} \right) \times 100$$

$$P_{\text{os}} = \min(25.0, ; R_{\text{os}} \times 5.0)$$

3. Excess Idling Penalty (P_{idle}):

$$\text{Idle Rate } (R_{\text{idle}}) = \left(\frac{\text{Idle Hours}}{D_{\text{exp}}} \right) \times 100 = \left(\frac{T_{\text{idle_sec}}/3600}{D_{\text{exp}}} \right) \times 100$$

$$P_{\text{idle}} = \min(25.0, ; R_{\text{idle}} \times 4.0)$$

4. Fuel Inefficiency Penalty (P_{eff}):

$$\text{Efficiency Ratio } (E_{\text{ratio}}) = \frac{\text{Actual Efficiency achieved}}{\text{Baseline Target Efficiency}}$$

$$P_{\text{eff}} = \begin{cases} 0.0, & \text{if } E_{\text{ratio}} \geq 1.0 \\ \min(25.0, ; (1.0 - E_{\text{ratio}}) \times 40.0), & \text{if } E_{\text{ratio}} < 1.0 \end{cases}$$

5. Rash Driving / Harsh Braking Penalty (P_{rash}):

$$\text{Rash Rate } (R_{\text{rash}}) = \left(\frac{N_{\text{harsh_events}}}{D_{\text{exp}}} \right) \times 100$$

$$P_{\text{rash}} = \min(25.0, ; R_{\text{rash}} \times 5.0)$$

6. Total Normalized Driver Score:

$$\text{Driver Score} = \max \left(0, ; \min \left(100, ; \text{round}(100 - P_{\text{os}} - P_{\text{idle}} - P_{\text{eff}} - P_{\text{rash}}) \right) \right)$$

B. Grade Classification Table

- **\$90 - 100\$ Points: Excellent** (Eligible for driver incentive bonuses)
- **\$75 - 89\$ Points: Good** (Standard compliant driver)
- **\$60 - 74\$ Points: Needs Improvement** (Targeted coaching recommended)
- **\$< 60\$ Points: Poor** (Safety risk flag triggered)

3.13 Normalized Vehicle Health Score Engine

The Vehicle Health Engine monitors mechanical wear, chassis vibration, degraded fuel efficiency, and operating alerts.

A. Mathematical Formula

Vehicle Health Score = $\max\left(0, \min\left(100, 100 - P_{\text{vib}} - P_{\text{eff_deg}} - P_{\text{alerts}} - P_{\text{fuel_drop}}\right)\right)$

Where:

1. Abnormal Vibration Penalty (P_{vib}):

$P_{\text{vib}} = \begin{cases} \min(25, \text{round}((\text{Vibration} - 5.0) \times 3.0)), & \text{if } \text{Vibration} > 5.0 \text{ g} \\ 0, & \text{otherwise} \end{cases}$

2. Efficiency Degradation Penalty ($P_{\text{eff_deg}}$):

$P_{\text{eff_deg}} = \begin{cases} \min(25, \text{round}((M_{\text{threshold}} - M_{\text{current}}) \times 5.0)), & \text{if } 0 < M_{\text{current}} < M_{\text{threshold}} \\ 0, & \text{otherwise} \end{cases}$

3. Mechanical Alerts Penalty (P_{alerts}):

$P_{\text{alerts}} = \min(25, N_{\text{mechanical_alerts}} \times 5.0)$

(Excludes pure GPS connectivity dropouts; focuses on engine, overheating, and sensor alerts).

4. Unprevented Fuel Abnormality Penalty ($P_{\text{fuel_drop}}$):

$P_{\text{fuel_drop}} = \min(25, N_{\text{fuel_loss_events}} \times 5.0)$

3.14 Audited Savings Wallet & 7-Day Baseline Learning Engine

To provide audit-grade accounting, DravYantra divides fleet fuel economics into **Three Strictly Isolated Buckets**:

AUDITED SAVINGS WALLET ARCHITECTURE	
1. Verified Net Savings (₹ / L)	Achieved when current efficiency > 7-day completed baseline efficiency.
2. Fuel Loss Prevented (₹ / L)	Direct loss prevented by immediate alert intervention (theft aborted, idle stopped).
3. Identified Fuel Waste (₹ / L)	Sum of Idling Loss + Overspeeding Loss + Unprevented Theft Loss.

Baseline Learning Lifecycle:

1. When a vehicle is onboarded, its `vehicle_baselines` status is set to `'collecting'`.
2. Telemetry is gathered for **\$7\text{ days}\$** or until **\$100\text{ km}\$** of driving history is accumulated.
3. Baseline efficiency is locked:

Baseline Efficiency (M_{baseline}) = $\frac{\sum \text{Distance}}{\sum \text{Fuel Consumed}}$

4. Once completed, verified savings are credited monthly to the organization's savings wallet.

3.15 Vehicle Lifetime Statistics Aggregation Engine

To prevent performance degradation from querying millions of historical records, the backend maintains cached lifetime metrics on the `vehicle_lifetime_stats` table.

- **Lifetime Odometer:** $\sum \text{distance}$ across all historical trips.
- **Lifetime Fuel Consumed:** $\sum \text{fuel_used}$.
- **Lifetime Financial Savings:** $\sum \text{money_saved}$.
- **Lifetime Carbon Avoided:** $\sum \text{co2_emitted_avoided}$.

4. DRAVYANTRA FLEET MANAGER DASHBOARD: EXHAUSTIVE FEATURE BREAKDOWN

The DravYantra Fleet Manager Dashboard is a command center providing fleet operators with complete oversight of their mobile assets.

DRAVYANTRA FLEET MANAGER DASHBOARD MAP	
[1. Dashboard Overview]	→ Fleet KPIs, Live Summary, Efficiency, Monthly Trends
[2. Live Fleet Tracking]	→ Real-Time GIS Map, Vehicle Badges, Speed, Heading
[3. Trips Management]	→ Trip Dispatch, Real-Time Progress, History, Overrides
[4. Fuel Intelligence]	→ Refills, Fuel Theft Logs, Tank Level Graph, Sensors
[5. Savings Wallet]	→ Verified ROI, Prevented Loss, Identified Waste Breakdown
[6. Carbon ESG Analytics]	→ CO2 Emitted, CO2 Avoided, Tree Equivalency, Reports
[7. Driver Scorecards]	→ Safety Rankings, Penalty Breakdown, Driver Profiles
[8. Vehicle Health]	→ Vibration Telemetry, Mechanical Health, Maintenance
[9. Alerts Watchtower]	→ Real-time Sound/Push Feed, Severity Categorization
[10. Reports & Exports]	→ Custom PDF / Excel Report Builder, Scheduled Emails
[11. Fleet Settings]	→ Threshold Calibration, Fuel Price, Notifications
[12. Support Helpdesk]	→ Ticket Creation, SLA Tracking, Priority Support

4.1 Executive Overview & Fleet KPI Dashboard

- **Top Metric Cards:**
 - **Active Vehicles Running / Idle / Offline:** Live status count of all fleet assets.
 - **Total Distance Traveled (km):** Filterable by Today, This Week, This Month, or Custom Date Range.
 - **Total Fuel Consumed (L) & Total Fuel Cost (₹):** Financial expenditure on diesel.
 - **Net Verified Savings (₹):** Total savings generated through eco-speeding.
 - **Total Identified Waste (₹):** Money lost due to excess idling, overspeeding, and fuel loss.
- **Visual Analytics Charts:**
 - **Fuel vs Loss Monthly Trend (Bar & Line Chart):** Monthly visual comparing fuel consumed vs money wasted.
 - **Fleet Status Donut Chart:** Breakdown of vehicle states (Moving, Idle, Parked, Disconnected).

4.2 Live Fleet GIS Tracking & Interactive Map

- **Interactive OpenStreetMap / Leaflet Engine:** Smooth vector map with clustered vehicle markers.
- **Dynamic Vehicle Marker Pins:** Color-coded pins (Green = Moving, Orange = Idling, Red = Overspeeding / Critical Alert, Gray = Engine Off / Parked).
- **Live Vehicle Detail Hover Card:** Displays real-time speed (km/h), remaining fuel (Liters & Percentage), driver name, current trip ID, and last GPS ping timestamp.
- **Geofencing Overlays:** Visual boundary polygons indicating hubs, customer drop zones, and restricted areas.

4.3 Trips Management, Dispatch & Route Analysis

- **Create & Dispatch Trip Modal:** Input origin, destination, assigned vehicle plate, driver name, expected distance, and cargo notes.
 - **Active Trips Monitor:** Real-time progress bar showing distance covered, live speed, fuel used, and live safety score.
 - **Historical Route Playback:** Breadcrumb trail visualization showing past route, speed variations, and event locations (e.g., where harsh braking occurred).
 - **Manual Trip Override:** Allows managers to close stale trips or adjust cargo status.
-

4.4 Fuel Intelligence, Theft Audits & Refills

- **Live Fuel Level Gauges:** Displays current fuel volume in liters for all active vehicles.
 - **Theft Incident Log:** Comprehensive table of detected fuel theft events with timestamps, fuel drop amounts, financial values, and location links.
 - **Refill Verification Registry:** Logs automated fuel refill events, verifying fuel volume added against fuel station receipts.
 - **Idle vs Moving Fuel Consumption Split:** Compares fuel burned while moving vs fuel wasted while stationary.
-

4.5 Audited Savings Wallet & ROI Analytics

- **Verified Net Savings Module:** Direct fuel savings credited based on actual vs baseline performance.
 - **Fuel Loss Prevented Ledger:** Documents losses averted through timely alerts (e.g., idle warnings acted upon).
 - **Identified Waste Breakdown:** Detailed breakdown of losses into Idling, Overspeeding, and Theft.
 - **Vehicle-by-Vehicle Savings Table:** Compares individual vehicle baseline efficiency against current efficiency.
-

4.6 Carbon ESG & Sustainability Reporting

- **Gross $\text{\$CO}_2$ Emitted (Metric Tonnes):** Total emissions calculated from fuel burned ($\text{\$2.68} \times \text{kg CO}_2 / \text{L}$).
 - **Net $\text{\$CO}_2$ Avoided (kg):** Greenhouse gas emissions prevented through eco-driving.
 - **Elemental Carbon ($\text{\$C}$) Mass Reduction (kg):** Molecular carbon savings ($\text{\$CO}_2 \times 12/44$).
 - **One-Click ESG Report Export:** Formatted PDF exports ready for corporate ESG sustainability filings.
-

4.7 Driver Scorecards, Safety Leaderboard & Behavior

- **Fleet Driver Leaderboard:** Ranks drivers from highest to lowest score based on exposure-normalized safety algorithms.
 - **Individual Driver Deep-Dive:**
 - Distance driven and trips completed.
 - Overspeeding events per $\text{\$100} \times \text{km}$.
 - Excess idling hours per $\text{\$100} \times \text{km}$.
 - Fuel efficiency vs baseline target.
 - Harsh braking and rash driving incidents.
 - **Coaching Recommendations:** Auto-generated behavioral suggestions based on penalty triggers.
-

4.8 Vehicle Health, Diagnostics & Maintenance Schedules

- **Vehicle Health Score (0–100):** Real-time mechanical status for every registered vehicle.
 - **Vibration & Tampering Telemetry:** Tracks chassis vibration in $\text{\$g}$ -force to detect suspension wear or unauthorized movement.
 - **Mileage Degradation Alerts:** Identifies vehicles consuming more fuel than standard thresholds.
 - **Service Logs & Maintenance Reminders:** Manage oil changes, tire rotations, and fitness certificate renewals.
-

4.9 Alerts, Tampering & Real-Time Incident Watchtower

- **Live Alert Feed:** Instant notifications for fuel theft, speed violations, long idling, harsh braking, and device tampering.
- **Severity Filters:** Categorized by Critical (Red), Warning (Amber), and Informational (Blue).

- **Audio & Visual Notifications:** Optional audio alerts for high-priority events.
- **Incident Resolution Tracking:** Managers can mark alerts as "Acknowledged", "Investigating", or "Resolved" with internal notes.

4.10 Automated Reports Generation & Data Export Hub

- **Multi-Format Export:** Generates reports in PDF, Microsoft Excel (.xlsx), and CSV formats.
- **Report Types:**
 - Trip Summary & Distance Log.
 - Fuel Consumption, Thefts & Refills Report.
 - Driver Safety & Compliance Audit.
 - Carbon ESG Sustainability Report.
 - Executive Financial ROI & Savings Summary.

4.11 Fleet Organization Settings & Threshold Calibration

- **Speed Limit Override:** Set custom fleet-wide speed limits (e.g., 70 km/h vs 80 km/h).
- **Fuel Theft Sensitivity:** Configure theft detection volume thresholds (0.5 L - 2.0 L).
- **Fuel Price Setting:** Maintain accurate local fuel costs (INR ₹ per liter).
- **Notification Routing:** Configure email and SMS alert recipients.

4.12 Support & In-App Helpdesk

- **Ticket Creation:** Submit technical issues directly to the DravYantra engineering team.
- **Priority Levels:** Low, Medium, High, and Urgent severity tags.
- **Ticket Tracking:** View status updates, assigned engineers, and resolution timestamps.

5. DRAVYANTRA SUPER-ADMIN DASHBOARD: PLATFORM OPERATIONS & ADMINISTRATION

The Super-Admin Dashboard is the central governance portal for DravYantra platform engineers and operations teams.

DRAVYANTRA SUPER-ADMIN CONSOLE MAP	
[1. Operations Center]	→ System Uptime, Telemetry Ingestion Rate, Active Tenants
[2. Organization Manager]	→ Fleet Onboarding Approvals, Suspension, Tier Limits
[3. Hardware Inventory]	→ Device Provisioning, IMEI Mapping, SIM Card Status
[4. Vehicle Global Index]	→ Cross-Tenant Vehicle Search, Plate Validation
[5. Global Trip Monitor]	→ Real-Time Platform Trips, Stalled Trip Resets
[6. Subscriptions Engine]	→ Plan Creation, License Billing, Invoicing, Gateways
[7. Global User & RBAC]	→ Super-Admins, Fleet Owners, Permissions Matrix
[8. Global Alerts Center]	→ Platform-Wide Tampering, Theft Clusters, Geo-Alerts
[9. Audit Trails & Logs]	→ Immutable Security Logs, Admin Actions, IP Tracking
[10. Support Resolution]	→ Global Helpdesk Triage, SLA Escalations, Admin Chat
[11. Recycle Bin Engine]	→ 30-Day Soft Deletion Recovery, Hard Purge Worker

5.1 Super-Admin Role & Global Responsibilities

The Super-Admin manages the entire platform infrastructure:

1. **Tenant Onboarding & Compliance:** Audits and approves fleet business registrations.
2. **Hardware & Telemetry Integrity:** Monitors device health and IoT communications.
3. **Billing & Subscriptions:** Manages commercial licensing and service agreements.

4. **Platform Security & Auditing:** Enforces data privacy and platform compliance.

5.2 Platform Operations Center & System Health Metrics

- **Global Ingestion Rate:** Displays real-time telemetry packets processed per second.
 - **Database Connection Pool Status:** Monitors active and idle PostgreSQL connections.
 - **Global Asset Summary:** Total registered organizations, active vehicles, online devices, and trips in progress.
-

5.3 Fleet Organizations & Onboarding Approval Lifecycle

- **Pending Onboarding Queue:** Review newly registered fleet accounts.
 - **Document Verification:** Inspect submitted company registration numbers, tax IDs, and contact info.
 - **Account Actions:** Approve, reject, suspend, or reactivate organization accounts.
 - **Fleet Size Limits:** Set maximum vehicle and user capacity per organization.
-

5.4 IoT Hardware & Telemetry Device Inventory Management

- **Hardware Provisioning:** Register new GPS/OBD-II hardware by IMEI, serial number, and model.
 - **Device-to-Vehicle Association:** Link devices to vehicle registration plates.
 - **Connection Health Tracking:** Identify devices that are Online, Offline, or experiencing Weak Signal.
 - **Over-The-Air (OTA) Firmware Management:** Monitor firmware versions and push remote configuration updates.
-

5.5 Global Vehicle & Asset Registry

- Search and filter vehicles across all organizations.
 - View vehicle details, assigned hardware, operational history, and active status.
 - Unlink hardware from decommissioned vehicles.
-

5.6 Global Trip Surveillance & Incident Investigation

- Live platform-wide map showing all trips in progress.
 - Filter trips by status (Running, Idle, Completed, Cancelled).
 - Emergency controls to pause or conclude stalled trips.
-

5.7 Subscription Plans, Billing & Commercial Engine

- **Tier Configuration:** Manage Starter, Growth, and Enterprise subscription plans.
 - **Billing Cycles:** Configure monthly, quarterly, or annual billing.
 - **Invoice Management:** Generate invoices, track payment receipts, and handle past-due accounts.
-

5.8 Global User Management & Role-Based Access Control (RBAC)

- **Role Hierarchy:**
 - **Super Admin** : Complete platform access.
 - **Platform Support** : Access to hardware diagnostics and support tickets.
 - **Fleet Owner / Manager** : Organization-level fleet management.
 - **Dispatcher** : Trip creation and live tracking access.
 - **Driver** : View personal trips and driver scores.
 - **User Actions:** Reset passwords, enforce Multi-Factor Authentication (MFA), and audit login histories.
-

5.9 Platform-Wide Security Alerts & Tampering Watchtower

- Real-time stream of all critical alerts across all tenants.
 - Detect coordinated fuel theft patterns across geographical regions.
 - Monitor device tampering and power-disconnect events.
-

5.10 Audit Trails, Security Logging & Compliance Tracker

- Immutable audit log recording every admin action (approvals, role modifications, configuration updates).
 - Records admin user ID, action type, IP address, timestamp, and affected entity.
-

5.11 Global Support Helpdesk & SLA Management

- Consolidated view of support tickets raised by all fleet managers.
 - Ticket assignment to support staff, status updates, and customer communication.
 - SLA monitoring with escalation triggers for overdue tickets.
-

5.12 Recycle Bin & 30-Day Soft Deletion Recovery Engine

- DravYantra implements a **30-day soft-deletion policy**:
 - When an organization or vehicle is deleted, it is moved to the Recycle Bin.
 - Data remains recoverable for 30 days.
 - An automated worker permanently purges records after the 30-day window expires.
 - Super-admins can restore accidentally deleted entities with a single click.
-

6. SUMMARY REFERENCE TABLE OF KEY BACKEND CONSTANTS & FORMULAS

Metric / Feature	Mathematical Formula	Constants / System Defaults
GPS Distance (km)	Haversine Formula	Earth Radius R = 6371.0 km, Drift Filter = 0.010 km
Dynamic Mileage (km/L)	Speed-Graduated Lookup	Eco (40-59 km/h) = 5.0 km/L; Baseline (60 km/h) = 4.0 km
Fuel Used (Liters)	Distance / Current_Mileage	Dynamic based on speed
Fuel Saved (Liters)	$\max(0, \text{Expected} - \text{Used})$	Expected = Distance / Baseline_Mileage
Money Saved (INR ₹)	Fuel_Saved x Fuel_Price	Default Fuel Price = ₹95.00 / Liter
Speeding Fuel Waste (L)	$\max(0, \text{Used} - \text{Expected})$	Applies when Speed > 69 km/h
Idling Money Waste (INR ₹)	Idle_Seconds x 1.70	₹1.70 / second (Base commercial idle burn rate)
Idling Fuel Waste (Liters)	Idle_Money_Waste / Price	Converted via current fuel price
Fuel Theft Detection	$\Delta_{\text{Fuel}} \geq 0.70 \text{ L}$	Window = 5 seconds, Power = OFF, Speed = 0 km/h
Fuel Refill Detection	$\Delta_{\text{Fuel}} \geq 5.00 \text{ L}$	Triggers automated refill audit event
Gross CO2 Emitted (kg)	Fuel_Used x 2.68	2.68 kg CO2 / Liter of Diesel
CO2 Avoided / Saved (kg)	Fuel_Saved x 2.68	2.68 kg CO2 / Liter of Diesel
Carbon (C) Mass Reduced (kg)	$\text{CO2_Avoided} \times (12 / 44)$	0.2727 Carbon-to-CO2 mass conversion ratio
Live Trip Score (0-100)	100 - Penalties	Overspeed: -2, Harsh Brake: -3, Rapid Accel: -2
Driver Scorecard (0-100)	Normalized per 100 km	Overspeed (Max 25), Idle (Max 25), Eff (Max 25), Rash (M
Vehicle Health Score (0-100)	100 - Mechanical Penalties	Vibration > 5g, Mileage Degradation, Alerts, Fuel Drops

Document Version: 2.4.0 (Enterprise Edition)

Platform: DravYantra Telematics & Fleet Intelligence Systems